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K-FIELD ENTROPY-PUMPED CURRENT FLOWS ON THE QUADRATIC PHASE-COST SURFACE

A foundational reference on exact scalar endpoint addressing, native multi-path entropy, curvature rectification, automatic phase-energy pumping, population current, and the locked Cell3 phase-cost geometry

K-Field endpoint geometry, phase cost, and entropy-pumped current

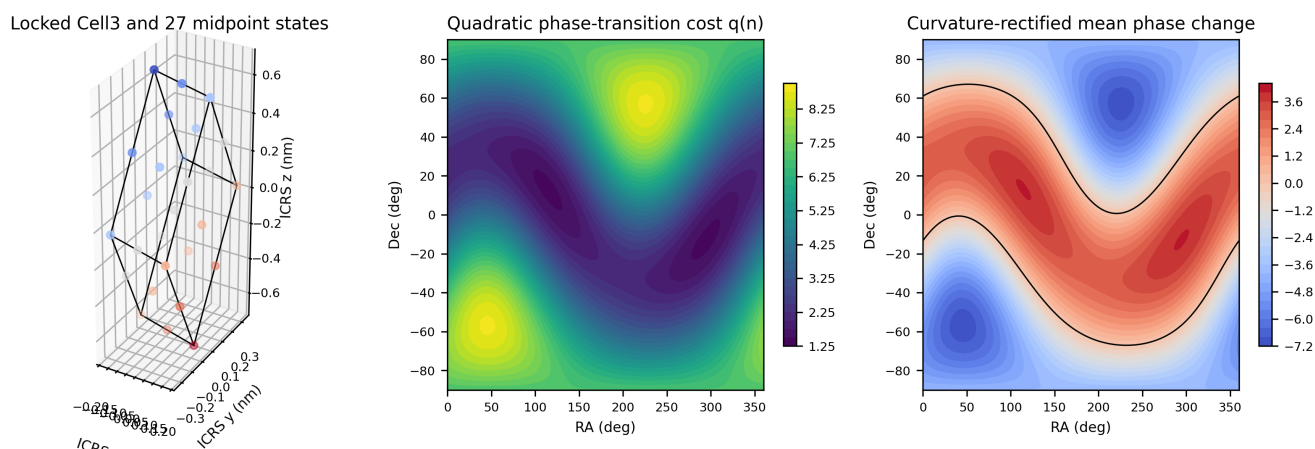


Figure Title. Title composite generated from the locked Cell3, the exact quadratic phase-cost matrix, and the exact isotropic curvature-rectification coefficient.

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Krytronx

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Phase-cost units	dimensionless q(n)
Entropy-penalty units	inverse square nanometres in q_S(n)
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Classification and use contract

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This reference presents the K-Field endpoint-address, path-entropy, quadratic phase-cost, curvature-rectification, and automatic-current structures as one internally self-sufficient mathematical system. Exact source values, exact enumerations, derived matrices, continuum limits, and measured curvature statistics are kept distinct by their declared definitions and units.

Contents

Document Control	2
Classification and use contract	2
Abstract	6
Foundational statement	6
1. Coordinate, dimensional, and state-space contract	6
1.1 ICRS physical frame	6
1.2 Distinct spaces and units	6
2. Locked Cell3 geometry, midpoint nodes, and boundary planes	7
2.1 Ordered basis and direct metric	7
2.2 The complete 27-node midpoint state set	8
2.3 Six primitive boundary planes	9
3. Lossless one-dimensional serialized spectral addressing	10
3.1 Central address theorem	10
3.2 Native scalar arithmetic	10
3.3 Endpoint certainty does not remove path uncertainty	11
4. Native multi-path propagation and entropy	11
4.1 Equal-weight primitive transition alphabet	11
4.2 Endpoint fibers	11
4.3 Exact entropy chain rule	12
5. Quadratic phase-transition cost surface	13
5.1 Directional cost field	13
5.2 Address-energy representation	13
6. Cost and entropy are complementary K-Field properties	14
6.1 The natural endpoint weight	14
6.2 Exact phase-selection boundary	14
7. Exact 10-event test of phase cost versus path entropy	14

7.1 Direct endpoint comparison	14
7.2 Interpretation of the near-zero correlation	15
8. The entropy-generated quadratic orientation surface	15
8.1 Large-event equal-weight limit	15
8.2 Transformation into physical ICRS space	15
9. Misalignment and competition between the two quadratic surfaces	16
9.1 Non-coaxial eigensystems	16
9.2 Phase-axis comparison	17
10. Effective directional population matrix	17
10.1 Combined large-event action	17
10.2 Crossover scales evaluated on the phase axes	17
11. Phase isolines, precession, and residence time	18
11.1 Cost-conserving orientation flow	18
11.2 Dwell-time weighting	19
12. Curvature organization beyond scalar cost and speed	19
13. Unified K-Field population measure	19
13.1 Three native weighting layers	19
13.2 Endpoint certainty plus propagation entropy	19
14. Symmetric path resolution on a curved cost surface	20
14.1 Equal path length does not imply equal energy change	20
15. Multidimensional Hessian-covariance law	21
15.1 Zero mean displacement and nonzero second moment	21
15.2 Exact quadratic contraction	21
15.3 Exact one-event Cell3 covariance	21
16. Exact curvature rectification on the quadratic orientation sphere	21
16.1 Equal geodesic paths	21
16.2 Spherical tangent Hessian	22
17. Soft, saddle, and hard phase axes	22

17.1 Eigenaxis directional coefficients	22
17.2 Saddle selection by path anisotropy	22
18. Isotropic path resolution and the neutral-current contour	23
18.1 Tangent-direction average	23
19. Entropy-pumped phase-energy and population currents	23
19.1 Local phase-energy pumping rate	23
19.2 Population current from the cost-entropy action	24
19.3 Coupling to cost-conserving precession	24
20. Central findings	25
21. New pathways to discovery and direct tests	25
21.1 Event-count scaling	25
21.2 Radius scaling	25
21.3 Controlled transition bias	25
21.4 Curvature-conditioned entropy test	25
21.5 Recursive Lucas and Supergroup extension	25
21.6 Curvature-reversal experiment	26
21.7 Saddle anisotropy experiment	26
21.8 Event-rate scaling	26
22. Reference formulas	26
Appendix A. Complete 27-node Cell3 ledger	27
Appendix B. Six boundary-plane membership ledger	28
Appendix C. Exact 10-event minimum-step shell ledger	28
Appendix D. Phase-cost decile and entropy ledger	28
Appendix E. Source and integrity ledger	29
Archival colophon	29

Abstract

The K-Field provides one explicit signed scalar address for every admissible location relative to a selected start point. The address is deterministic and lossless: it decodes to one integer lattice vector and therefore to one physical Cell3 displacement, one laminar grade, one polarity, and one recursive address. At the same time, an endpoint generally has many equal-weight admissible histories. The endpoint is certain while the route is uncertain. This route multiplicity produces a native combinatorial entropy that does not arise from measurement error or incomplete state description.

The same corpus defines a quadratic phase-transition cost surface on projective orientation space. For a unit ICRS direction $\mathbf{n}$, $q(\mathbf{n}) = n^T \mathbf{G}_{\text{phase}} \mathbf{n}$. The soft, saddle, and hard eigenaxes give the stationary orientations and the anisotropic cost spread. The cost surface and the path entropy are separate native properties. Cost assigns directional resistance; entropy counts how many admissible histories feed an explicitly addressed endpoint.

Combining these structures gives the exact endpoint weight $\Omega_N(\mathbf{k}) \exp[-\beta_K q(\mathbf{k})]$ and the effective action $\beta_K q(\mathbf{k}) - \ln \Omega_N(\mathbf{k})$. Equal forward and reverse path resolutions cancel the first-order phase-cost slope but do not cancel the second-order curvature. The exact local mean is one half of the contraction between the phase-cost Hessian and the path covariance. On the unit orientation sphere this produces a signed automatic phase-energy pumping field with a neutral contour at $q = \text{tr}(\mathbf{G}_{\text{phase}})/3$. In the large-event limit, equal-weight path multiplicity also generates a second quadratic orientation surface $q_S(\mathbf{n}) = n^T \mathbf{G}_S \mathbf{n}$, where $\mathbf{G}_S = \mathbf{B}^T(-\mathbf{T})\mathbf{B}(-\mathbf{1})$. The complete K-Field flow is therefore organized by path multiplicity, curvature-rectified phase-energy pumping, the cost-entropy action, and precessional residence.

Foundational statement

The K-Field state space is simultaneously explicit and statistical. Every endpoint has one exact scalar address, while the set of equal-weight paths reaching that endpoint may be highly degenerate. Entropy is therefore an intrinsic property of K-Field propagation. Symmetric path uncertainty has zero first moment but nonzero covariance. The local curvature of the quadratic phase-transition cost surface rectifies that covariance into a nonzero mean phase-energy change and therefore an automatic current. The resulting cost-entropy gradient organizes endpoint population together with precessional residence and curvature focusing.

1. Coordinate, dimensional, and state-space contract

1.1 ICRS physical frame

All physical vectors are expressed as Cartesian columns in the International Celestial Reference System. The x axis points toward right ascension 0 degrees and declination 0 degrees. The y axis points toward right ascension 90 degrees and declination 0 degrees. The z axis points toward the north celestial pole. This fixed frame prevents hidden rotations or relabeling when the direct Cell3 geometry, the phase-cost surface, and the entropy surface are compared.

1.2 Distinct spaces and units

Object	Definition	Units and role
Address $\mathbf{k}$	integer triple in $\mathbb{Z}^3$	exact relational location relative to a start point
Physical displacement $\mathbf{r}$	$\mathbf{r} = \mathbf{B}\mathbf{k}$	nm; direct Cell3 geometry
Unit direction $\mathbf{n}$	$\mathbf{n} = \mathbf{r}/\ \mathbf{r}\ $	dimensionless projective orientation
Serialized address ρ	$\rho(\mathbf{k}) = \sigma(\mathbf{k}) \sqrt{k^T \mathbf{G} \mathbf{k}}$	one signed scalar that uniquely identifies $\mathbf{k}$
Phase cost q	$q(\mathbf{n}) = n^T \mathbf{G}_{\text{phase}} \mathbf{n}$	dimensionless directional transition cost
Path multiplicity Ω_N	number of admissible N-event histories reaching $\mathbf{k}$	integer degeneracy
Path entropy S_N	$S_N = -k_B \ln \Omega_N$	entropy associated with route uncertainty
Entropy penalty q_S	$q_S(\mathbf{n}) = n^T \mathbf{G}_S \mathbf{n}$	nm^{-2} ; directional suppression of path multiplicity at fixed radius

The address, the physical displacement, the orientation, the cost, and the entropy are related but are not interchangeable. The signed scalar determines the endpoint. Path multiplicity describes the number of admissible histories compatible with that endpoint. Phase cost evaluates the endpoint direction. The unit declarations prevent a scalar cost from being added directly to an entropy term until

a conversion coefficient is supplied.

2. Locked Cell3 geometry, midpoint nodes, and boundary planes

2.1 Ordered basis and direct metric

The physical Cell3 is generated by the ordered basis $B=[a\ b\ c]$. The column order is structural because it fixes the three primitive transition families and their direct-reciprocal correspondence. The direct metric is $G=B^TB$, and the physical cell volume is $|\det B|$.

```
B (nm) =
[ 0.327501242069651    0.035330775844051   -0.010129317626665]
[ -0.054897007886313    0.207319850441498   -0.389787702363804]
[ -0.037487502896305   -0.547401085391957   -0.689734408362922]

G = B^T B (nm^2) =
[ 0.111676057905435    0.020710333281606    0.043937235097789]
[ 0.020710333281606    0.343877732397120    0.296392758961523]
[ 0.043937235097789    0.296392758961523    0.627770610069385]
```

Quantity	Value
a	0.334179679073151 nm
b	0.586410890414836 nm
c	0.792319765037693 nm
A_ab	0.194869165469900 nm^2
A_ac	0.261106235799531 nm^2
A_bc	0.357809539128562 nm^2
Volume	0.117914891916699 nm^3

Dimensionally correct Cell3: 27 nodes and six primitive boundary planes

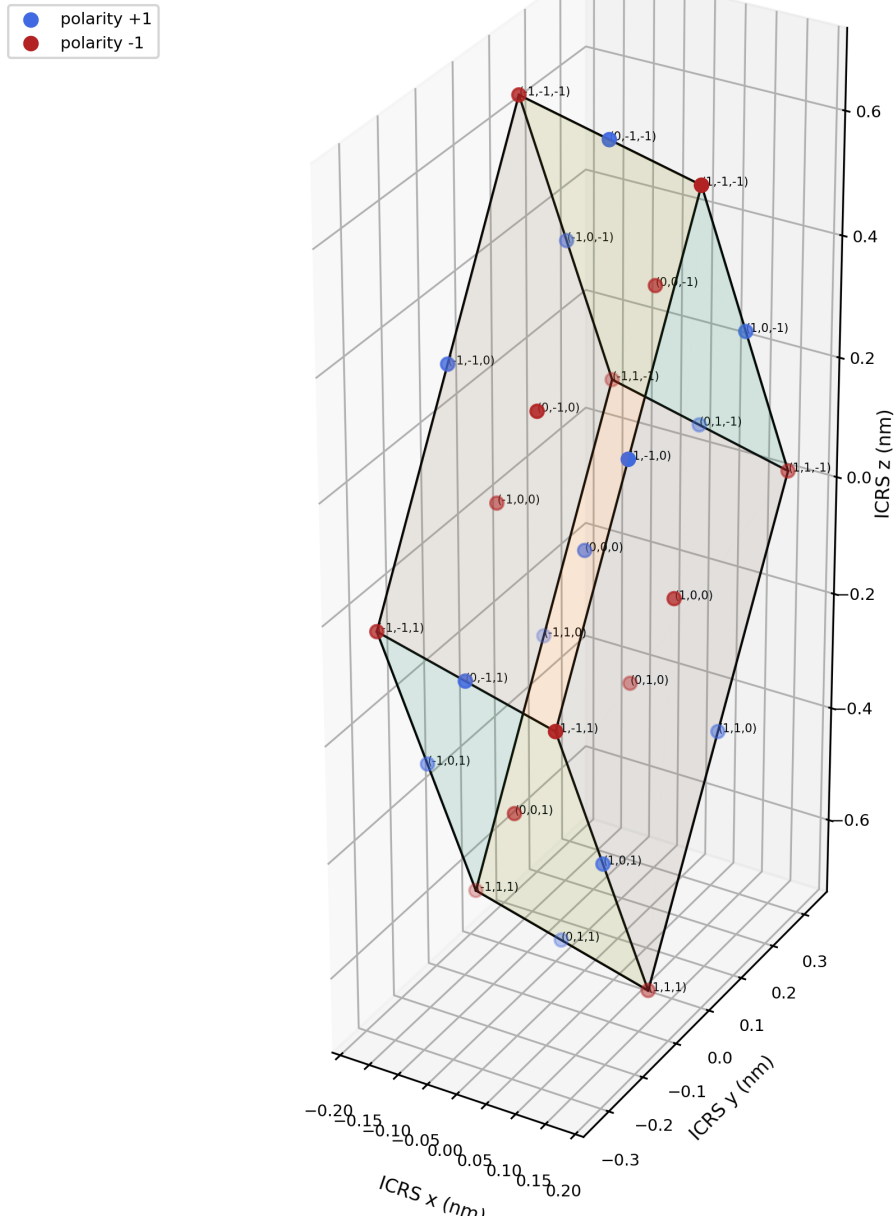


Figure 1. Dimensionally correct ICRS Cell3 with all 27 midpoint nodes, exact address labels, checkerboard polarity, and all six signed primitive boundary planes.

2.2 The complete 27-node midpoint state set

The midpoint address set is $L=\{-1,0,1\}^3$. Each node has physical position $p(L)=BL/2$. The laminar grade is $\lambda=i+j+k$ and checkerboard polarity is $\chi=(-1)^\lambda$. This creates seven sheets with populations 1, 3, 6, 7, 6, 3, and 1. The node kind is determined by the number of nonzero address components: three gives a corner, two an edge center, one a face center, and zero the body center.

Seven-sheet Cell3 laminar stack: populations 1, 3, 6, 7, 6, 3, 1

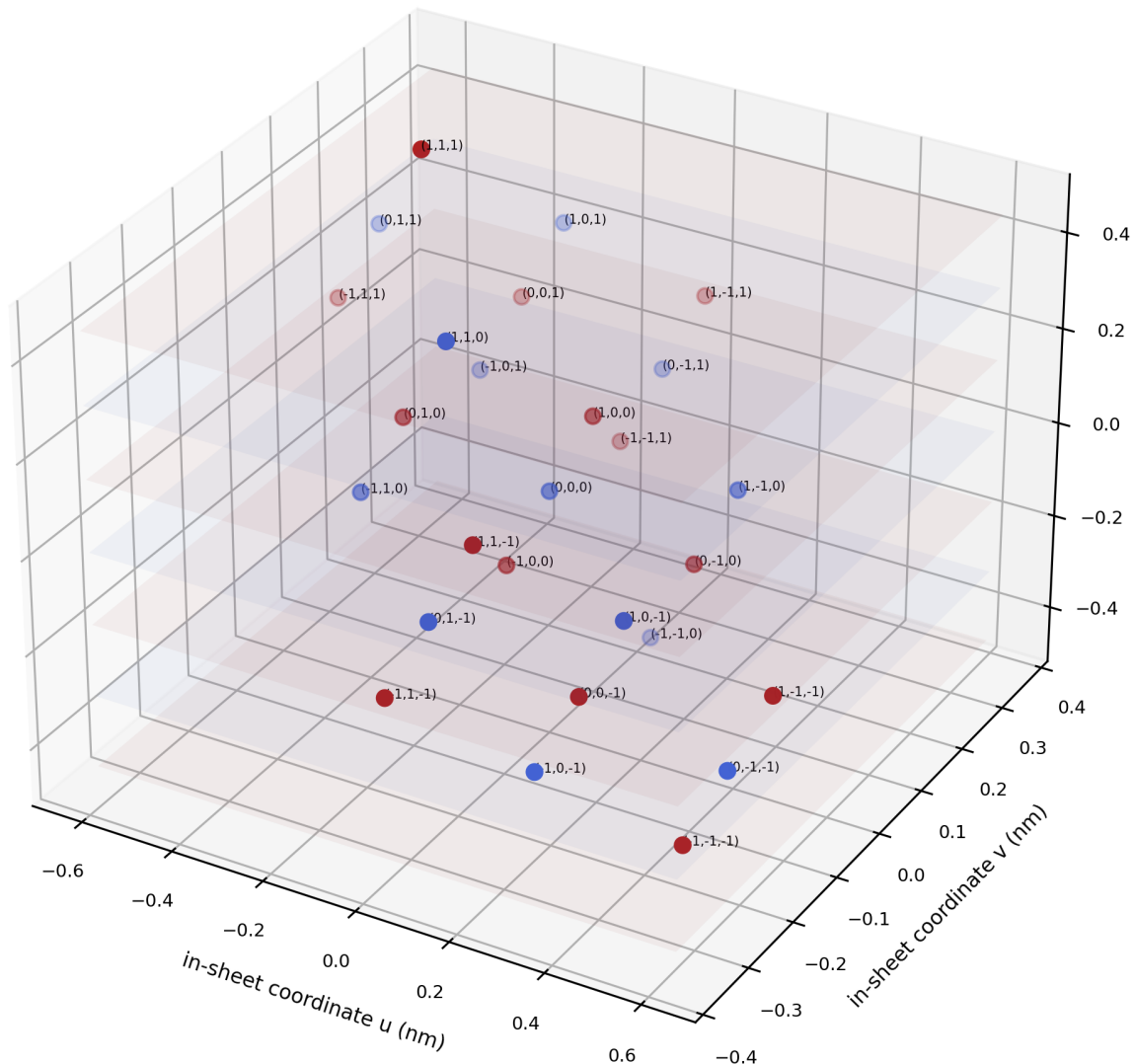


Figure 2. The exact seven-sheet decomposition of the 27 midpoint states. Every sheet has one polarity; each address is labeled.

2.3 Six primitive boundary planes

Each primitive face family has two signed boundary planes. Fixing address coefficient $a=+1$ or $a=-1$ selects the two bc planes; fixing b selects the ac planes; fixing c selects the ab planes. Every plane contains exactly nine midpoint nodes. The face areas and opposite spacings are generated from the same ordered basis and preserve the full oblique geometry.

All six primitive boundary planes and all nine midpoint nodes per plane

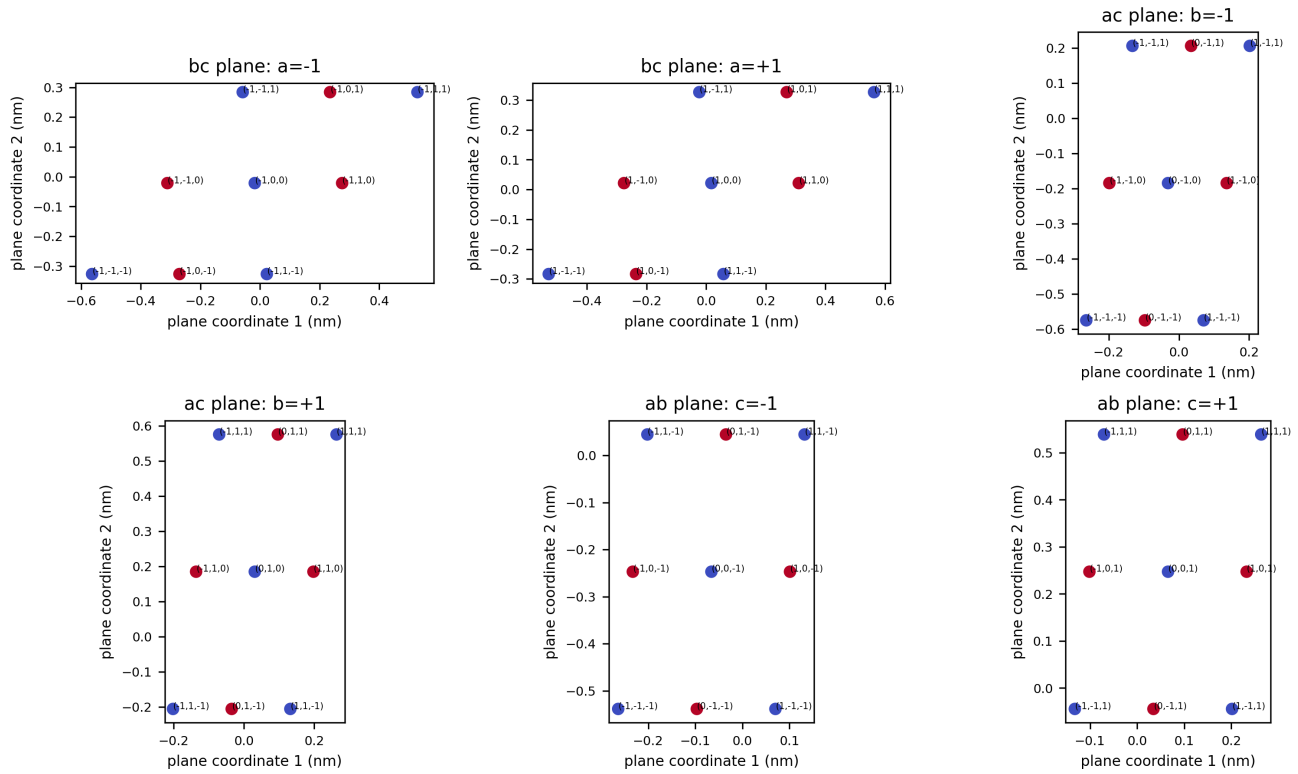


Figure 3. Six-panel atlas of the signed primitive boundary planes. Each panel preserves true in-plane metric and labels all nine member nodes.

3. Lossless one-dimensional serialized spectral addressing

3.1 Central address theorem

Let the relational lattice be $\Lambda = B \mathbb{Z}^3$ and define the direct quadratic shell $Q(k) = k^T G k$. The signed scalar address is $\rho(k) = \sigma(k) \sqrt{Q(k)}$, where the sign function supplies directed antipodal ordering. Under the corpus injectivity condition $Q(k) = Q(l)$ only when $l = +k$ or $l = -k$, the signed coordinate is a bijection: $\rho(k) = \rho(l)$ implies $k = l$.

$$\begin{aligned} \Lambda &= B \mathbb{Z}^3 \\ Q(k) &= k^T G k \\ \rho(k) &= \sigma(k) \sqrt{Q(k)} \\ \rho(k) &= \rho(l) \Rightarrow k = l \end{aligned}$$

One scalar therefore carries the complete directed endpoint address. Decoding ρ recovers k , and k recovers the physical displacement Bk , laminar grade, polarity, projective channel, and balanced-ternary recursive digits. The scalar coordinate serializes the hidden rank-three relational geometry rather than collapsing it.

3.2 Native scalar arithmetic

Addition and subtraction are performed by decoding, operating in the integer module, and re-encoding. This preserves the relational geometry even though the explicit coordinate is one-dimensional.

$$\begin{aligned} x (+) y &= \rho[\rho^{-1}(x) + \rho^{-1}(y)] \\ x (-) y &= \rho[\rho^{-1}(x) - \rho^{-1}(y)] \\ _G &= [x^2 + y^2 - (x (-) y)^2] / 2 \end{aligned}$$

3.3 Endpoint certainty does not remove path uncertainty

A unique endpoint address does not imply a unique history. The documented address (4,-2,2) has one scalar endpoint and 420 shortest paths. The paths consist of four +a events, two -b events, and two +c events in every distinct ordering. The multiplicity is $8!/(4!2!2!)=420$.

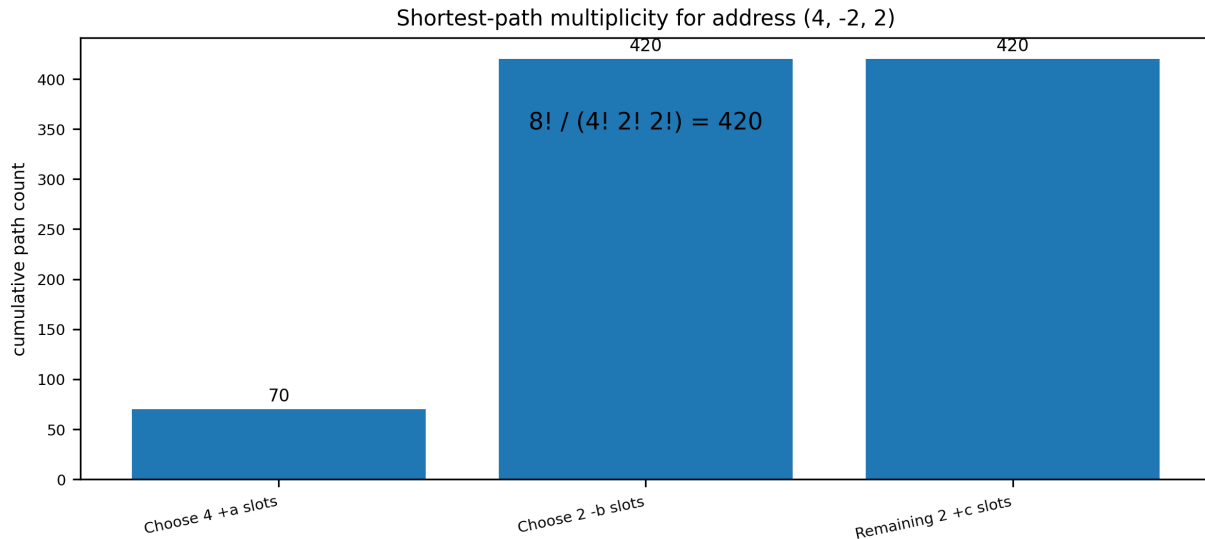


Figure 4. The documented serialized endpoint (4,-2,2) has 420 equal-length shortest paths, obtained by exact multinomial ordering.

4. Native multi-path propagation and entropy

4.1 Equal-weight primitive transition alphabet

The exact 10-event propagation experiment begins at Cell3 corner address (1,1,1) and uses the six directed primitive moves +a, -a, +b, -b, +c, and -c. Each event has probability 1/6. The tiling is unbounded. Exact dynamic programming enumerates all $6^{10}=60,466,176$ ordered histories without Monte Carlo sampling.

4.2 Endpoint fibers

For N events, $\Gamma_N(k)$ is the set of admissible ordered histories ending at address k, and $\Omega_N(k)=|\Gamma_N(k)|$ is the fiber multiplicity. Equal-weight histories give conditional path probability $1/\Omega_N(k)$ after the endpoint is known.

```

Gamma_N(k) = {ordered admissible N-event paths ending at k}
Omega_N(k) = |Gamma_N(k)|
P(path | k, N) = 1 / Omega_N(k)
S_N(k) = k_B ln Omega_N(k)

```

This entropy is not an uncertainty in the endpoint. It is the exact logarithmic size of the path fiber above an explicit endpoint. The K-Field is therefore deterministic in address and non-deterministic in propagation.

4.3 Exact entropy chain rule

Entropy quantity	Exact value
History entropy	17.917594692280549 nats
Endpoint entropy	5.364728188272016 nats
Mean conditional fiber entropy	12.552866504008534 nats
Chain-rule residual	-1.776e-15 nats
Reachable endpoint count	891
Effective endpoint count exp(H)	213.733131612919
Effective fiber multiplicity exp(H)	282905.020591319306

The fraction of total propagation entropy remaining after the endpoint is known is 70.058882 percent. Path uncertainty is therefore the dominant part of the 10-event information budget.

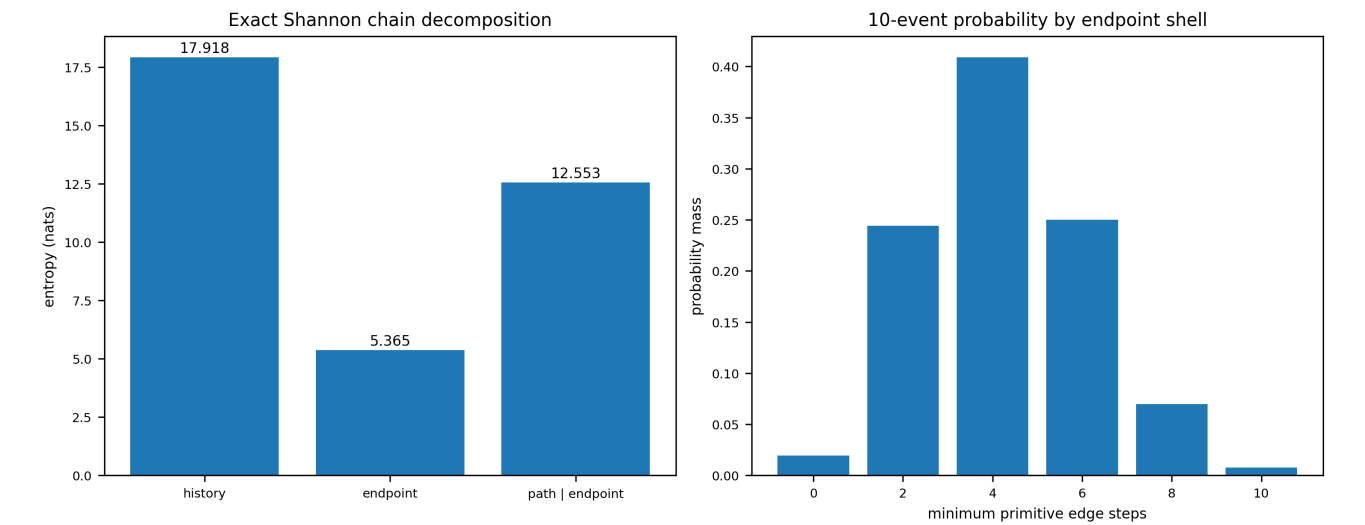


Figure 5. The 10-event history entropy separates exactly into endpoint entropy and mean conditional path-fiber entropy. The right panel shows probability by minimum-step shell.

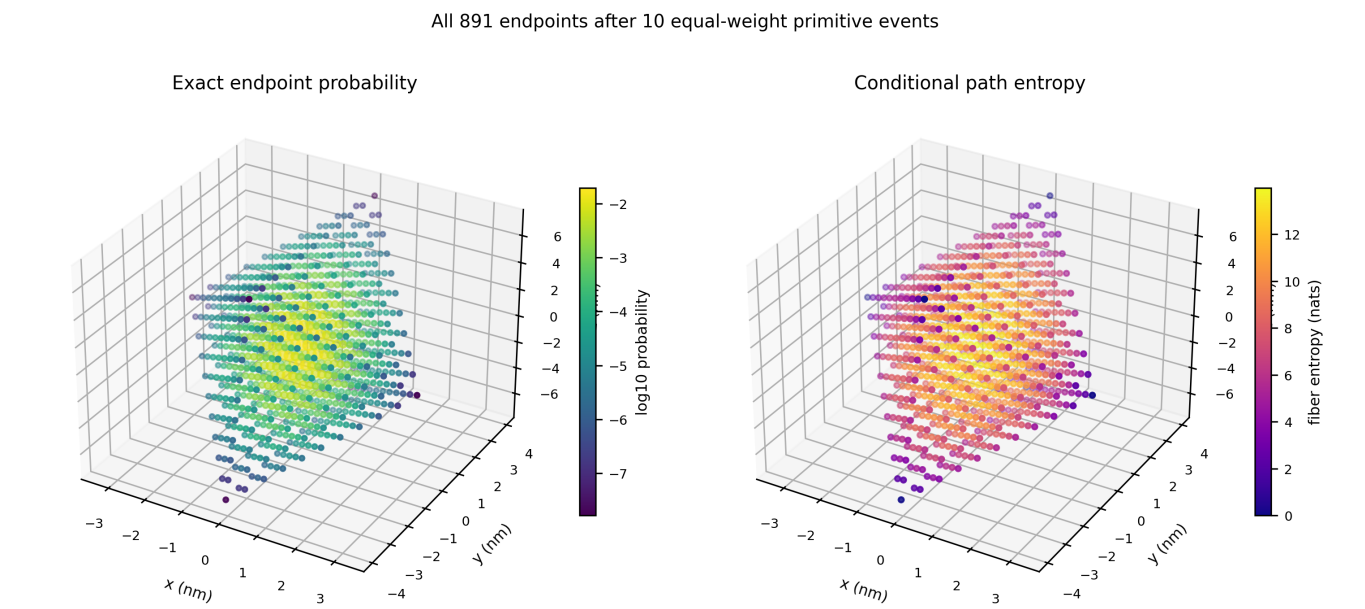


Figure 6. Exact dynamic-programming endpoint distribution for all 60,466,176 ordered histories. Position is explicit; color records probability or path-fiber entropy.

5. Quadratic phase-transition cost surface

5.1 Directional cost field

The K-Field directional cost is the positive-definite quadratic form $q(n)=n^T G_{\text{phase}} n$ restricted to unit ICRS directions. Because n and $-n$ have the same cost, the natural orientation space is projective. The surface is a scalar field on orientation space, not a physical ellipsoid in position space.

```
G_phase =
[ 2.976410637800  1.243150779500 -2.080360035400]
[ 1.243150779500  2.592228663500 -2.317883993700]
[-2.080360035400 -2.317883993700  6.773570913300]
```

$q(n) = n^T G_{\text{phase}} n, \quad ||n|| = 1$

Axis	q eigenvalue	RA (deg)	Dec (deg)	q_S at phase axis (nm ⁻²)
soft	1.429328753487	114.813083651	12.969895704	5.738931542435
saddle	2.112545026447	197.222212119	-29.836687522	7.103047066506
hard	8.800336434665	225.519007199	56.919969623	4.000616506795

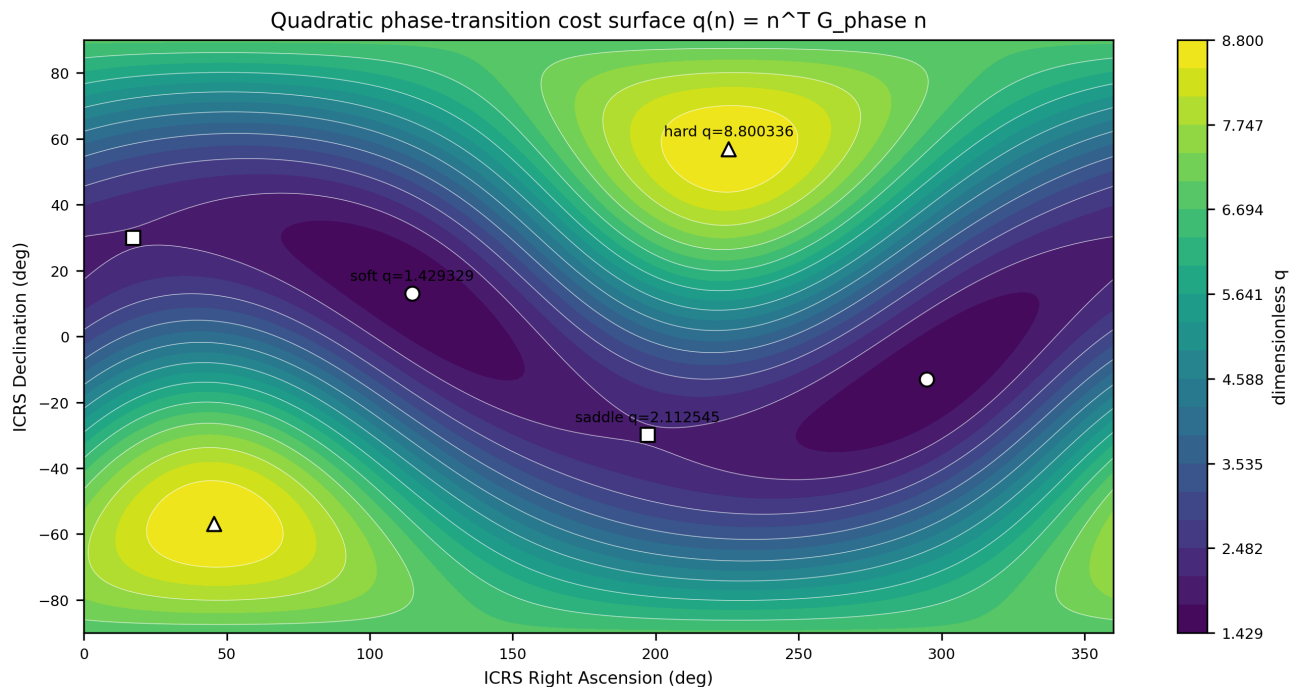


Figure 7. Exact ICRS quadratic phase-transition cost surface with soft, saddle, and hard stationary directions and their antipodes.

5.2 Address-energy representation

The phase matrix is generated from the local address-energy coupling matrix W through the exact identity $G_{\text{phase}}=4B^T(-T)WB^T(-1)$. This relation shows that the directional cost in ICRS space is the physical image of a quadratic address-space form.

```
G_phase = 4 B^T(-T) W B^T(-1)
q(k) = [k^T B^T G_phase B k] / [k^T G k]
      = 4 [k^T W k] / [k^T G k]
```

Because k is decoded from ρ , the phase cost is an explicit scalar-address observable: $q(x)=q[\rho^T(-1)(x)]$. No additional orientation state is needed.

6. Cost and entropy are complementary K-Field properties

6.1 The natural endpoint weight

Equal-weight path multiplicity and directional phase cost combine multiplicatively. For an endpoint k reached in N events, the unnormalized state weight is $\Omega_N(k) \exp[-\beta_K q(k)]$. The coefficient β_K converts the dimensionless phase cost into logarithmic statistical weight and is common to the complete ensemble.

$$\begin{aligned} w_N(k) &= \Omega_N(k) \exp[-\beta_K q(k)] \\ A_N(k) &= \beta_K q(k) - \ln \Omega_N(k) \end{aligned}$$

The effective action A_N is lower when the phase cost is lower or when the number of admissible paths is larger. A high-cost endpoint can dominate if it has enough path multiplicity. A low-cost endpoint can remain weakly occupied if only a small path fiber reaches it.

6.2 Exact phase-selection boundary

Two endpoint families i and j have equal effective weight when their phase-cost difference equals their entropy difference in logarithmic units.

$$\begin{aligned} A_i &= A_j \\ \beta_K (q_i - q_j) &= \ln[\Omega_i / \Omega_j] \\ &= (S_i - S_j)/k_B \end{aligned}$$

This equality is the exact cost-entropy transition condition. It defines where a preferred endpoint family changes as phase weighting, propagation depth, or path multiplicity changes.

7. Exact 10-event test of phase cost versus path entropy

7.1 Direct endpoint comparison

The phase cost was evaluated for every nonzero endpoint in the exact 10-event destination registry. The endpoint unit direction is $\mathbf{n} = \mathbf{B}_k / \|\mathbf{B}_k\|$, and $q = \mathbf{n}^T \mathbf{G}_{\text{phase}} \mathbf{n}$. This produces 890 nonzero endpoint comparisons with no sampling.

Statistic	Value
Nonzero endpoints	890
Equal-endpoint corr(q , fiber entropy)	-0.034449777988
Probability-weighted corr(q , fiber entropy)	0.001368924044
Probability-weighted mean q	4.696481203521
Probability-weighted mean fiber entropy	12.524749223618 nats
Unweighted q regression coefficient after distance and step count	0.018379535576
Weighted q regression coefficient after distance and step count	0.004413309583

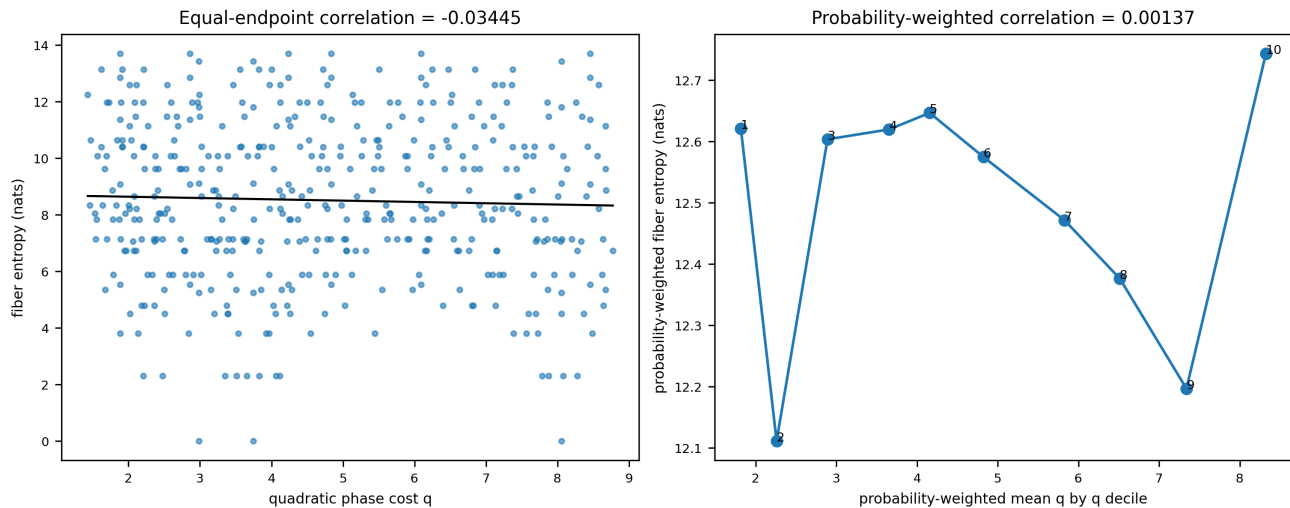


Figure 8. Exact 10-event comparison. The equal-weight propagation baseline carries substantial path entropy but no material correlation with the independent phase-cost surface.

7.2 Interpretation of the near-zero correlation

The exact unbiased propagation experiment assigns probability 1/6 to each primitive transition and does not use G_{phase} when selecting events. Its path entropy therefore measures combinatorial route degeneracy only. The near-zero correlation is not a failure of either structure. It demonstrates that native path entropy and the quadratic phase-cost surface are independent inputs to the complete K-Field population law.

If the cost surface had already been implicit in the equal-weight path count, a strong correlation would appear without any cost weighting. It does not. The phase surface must act on the entropy-bearing endpoint ensemble rather than being identified with the entropy itself.

8. The entropy-generated quadratic orientation surface

8.1 Large-event equal-weight limit

For the six equal-probability primitive moves, the endpoint distribution approaches a three-dimensional Gaussian in address space. The leading exponent is $-(3/2N)k^{\wedge}T$ k. The exact finite-N distribution retains integer and parity structure, while the Gaussian gives the continuous directional law.

$$P_N(k) = \Omega_N(k) / 6^{\wedge}N$$

$$P_N(k) \text{ proportional to } \exp[-(3/(2N)) k^{\wedge}T k]$$

8.2 Transformation into physical ICRS space

At fixed physical radius R , write $r=Rn$ and $k=B^{\wedge}(-1)r$. Then $k^{\wedge}T k=R^{\wedge}2 n^{\wedge}T B^{\wedge}(-T)B^{\wedge}(-1)n$. This produces a second quadratic surface on orientation space.

$$G_S = B^{\wedge}(-T) B^{\wedge}(-1)$$

$$q_S(n) = n^{\wedge}T G_S n$$

$$\ln \Omega_N(R, n) = \text{constant} - [3R^{\wedge}2/(2N)] q_S(n) + \text{higher-order terms}$$

$$G_S \text{ (nm}^{\wedge}-2) =$$

9.275643997185	0.095769243769	0.274708806720
0.095769243769	6.024094143790	-1.217784618059
0.274708806720	-1.217784618059	1.542856974761

Axis	q_S eigenvalue (nm ⁻²)	RA (deg)	Dec (deg)
soft	1.222872851451	98.306312947	75.582702912
saddle	6.333432713989	270.502388945	14.289263978

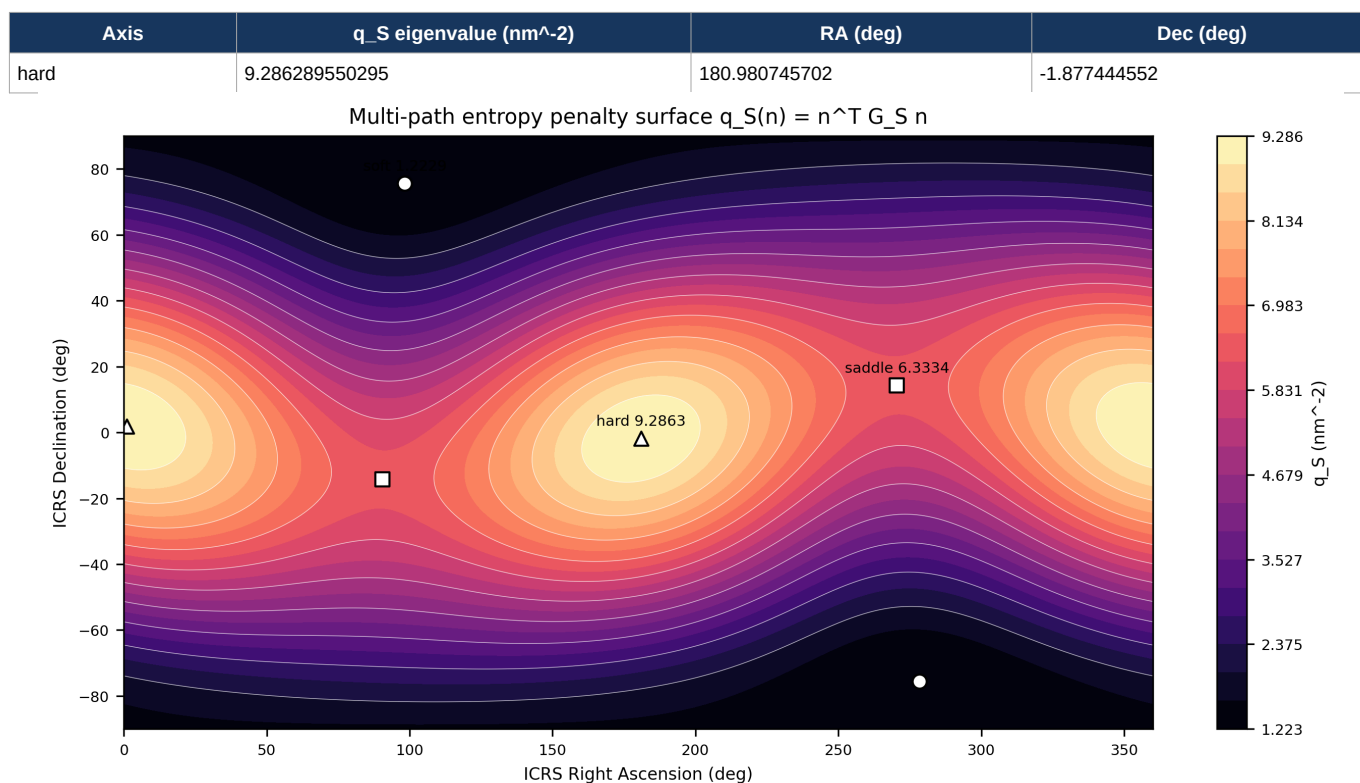


Figure 9. Derived physical-orientation penalty associated with equal-weight address-space diffusion. Lower q_S means greater path multiplicity at fixed physical radius.

The entropy surface is anisotropic in physical space even though the six moves are equiprobable in address space. The anisotropy arises because B maps equal address increments into unequal oblique physical vectors.

9. Misalignment and competition between the two quadratic surfaces

9.1 Non-coaxial eigensystems

G_phase and G_S do not share eigenvectors. The phase axes and entropy axes occupy different ICRS directions. Therefore the combined effective orientation cannot be reduced to rescaling one fixed ellipsoid. Changing the entropy-to-cost ratio rotates the effective eigenaxes.

Phase axis / Entropy axis	E-soft	E-saddle	E-hard
P-soft	63.256214 deg	36.369845 deg	67.282432 deg
P-saddle	58.980169 deg	83.163430 deg	31.931459 deg
P-hard	43.169375 deg	54.484404 deg	68.814817 deg

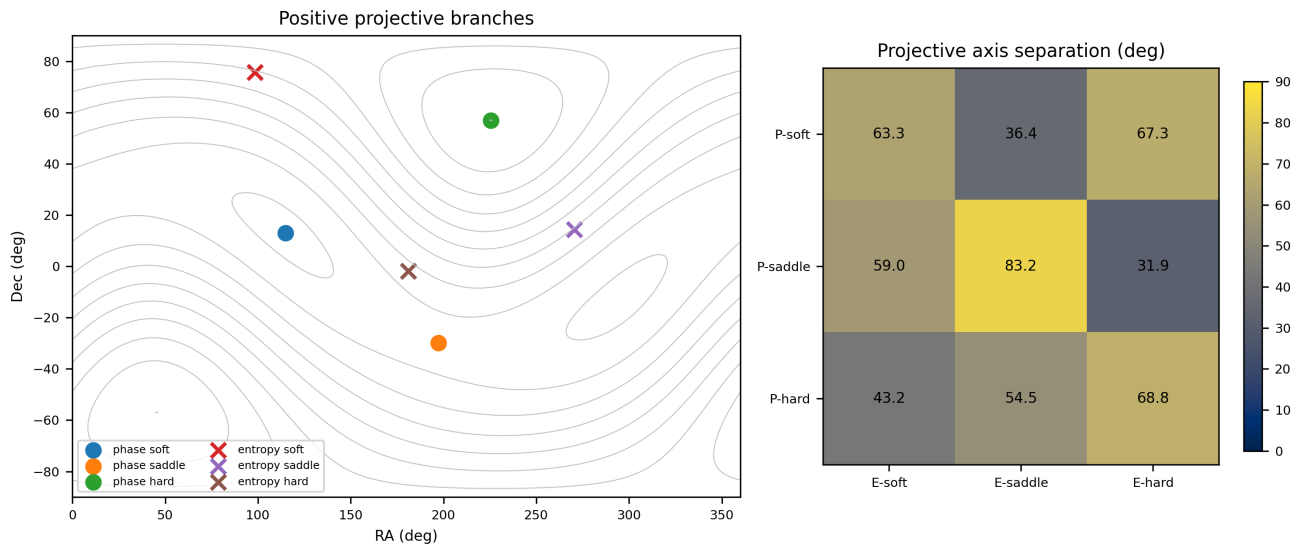


Figure 10. Comparison of the phase-cost and entropy-penalty eigensystems. The two quadratic surfaces are not coaxial.

9.2 Phase-axis comparison

Phase axis	q	q _S (nm ⁻²)	Cost role	Entropy role at fixed R
soft	1.429328753	5.738931542	minimum phase cost	intermediate entropy accessibility
saddle	2.112545026	7.103047067	intermediate phase cost	largest penalty among phase axes
hard	8.800336435	4.000616507	maximum phase cost	smallest penalty among phase axes

The phase-hard axis is strongly penalized by q but is comparatively favored by path entropy at fixed physical radius because its q_S value is lower than the q_S values at the phase-soft and phase-saddle axes. This establishes a direct competition: phase cost suppresses the hard direction while multi-path entropy partly restores it.

10. Effective directional population matrix

10.1 Combined large-event action

Substituting the asymptotic path multiplicity into the cost-entropy action gives one quadratic effective orientation matrix.

$$A_N(R, n) = n^T [\beta_K G_{\text{phase}} + 3R^2 G_S / (2N)] n + \text{constant}$$

$$G_{\text{eff}} = \beta_K G_{\text{phase}} + 3R^2 G_S / (2N)$$

Using $G_{\text{phase}} = 4B^{(-T)}WB^{(-1)}$ and $G_S = B^{(-T)}IB^{(-1)}$, the direct address-space form is

$$G_{\text{eff}} = B^{(-T)} [4 \beta_K W + 3R^2 I / (2N)] B^{(-1)}$$

The matrix W supplies phase-transition anisotropy. The identity matrix supplies equal-weight address-space diffusion. $B^{(-1)}$ maps both structures into physical ICRS orientation space. The ratio $c = [3R^2 / (2N)] / \beta_K$ has units of nm² and controls the competition.

10.2 Crossover scales evaluated on the phase axes

Axis pair	Delta q	Delta q _S (nm ⁻²)	crossover c (nm ²)
soft vs saddle	0.683216272960	1.364115524071	none for positive c
soft vs hard	7.371007681178	-1.738315035640	4.240317508651
saddle vs hard	6.687791408218	-3.102430559711	2.155661917165

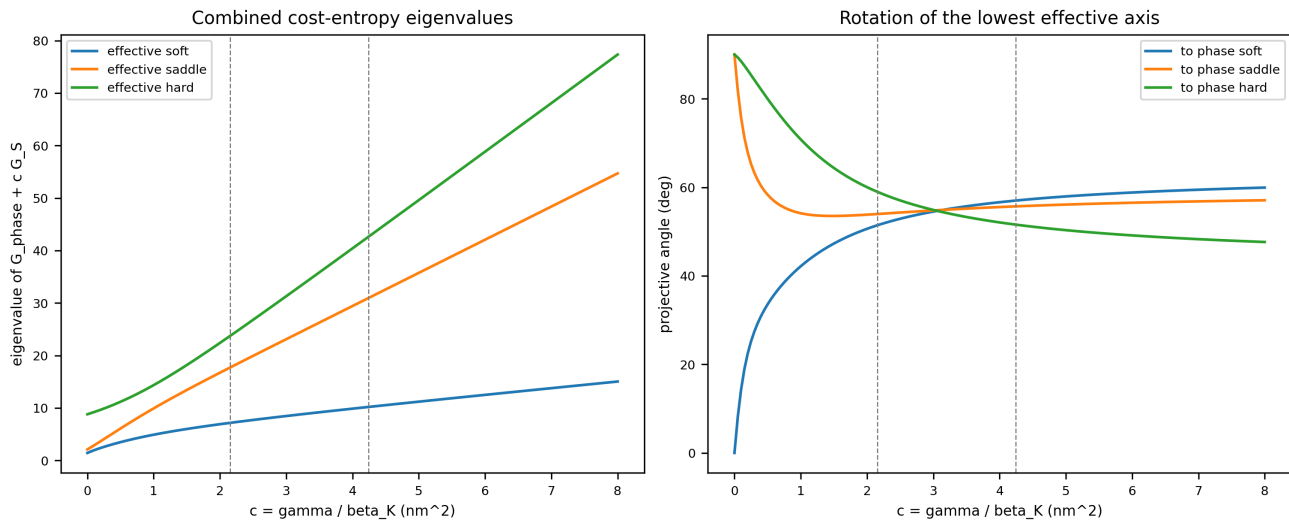


Figure 11. Parameter-free matrix sweep in $c = \gamma/\beta_K$ using the locked G_{phase} and G_S matrices. It shows how the preferred effective direction rotates as entropy weighting increases.

Because the matrices do not commute, the full effective eigenvectors rotate continuously rather than simply exchanging the original phase axes. The sweep is generated directly from the two locked matrices and requires no fitted endpoint coefficients.

11. Phase isolines, precession, and residence time

11.1 Cost-conserving orientation flow

For fixed angular-momentum magnitude L , the K-Field torque associated with the quadratic cost is tangent to the unit sphere. The orientation equation is $\dot{n} = -(2/L)n \times G_{\text{phase}} n$. The cross product preserves $\|n\|$ and gives $dq/dt=0$. An undamped trajectory remains on one phase-cost isoline.

$$\begin{aligned} \tau_K &= -2 n \times G_{\text{phase}} n \\ \dot{n} &= -(2/L) n \times G_{\text{phase}} n \\ dq/dt &= 0 \\ \| \dot{n} \| &= (2/L) \| n \times G_{\text{phase}} n \| \end{aligned}$$

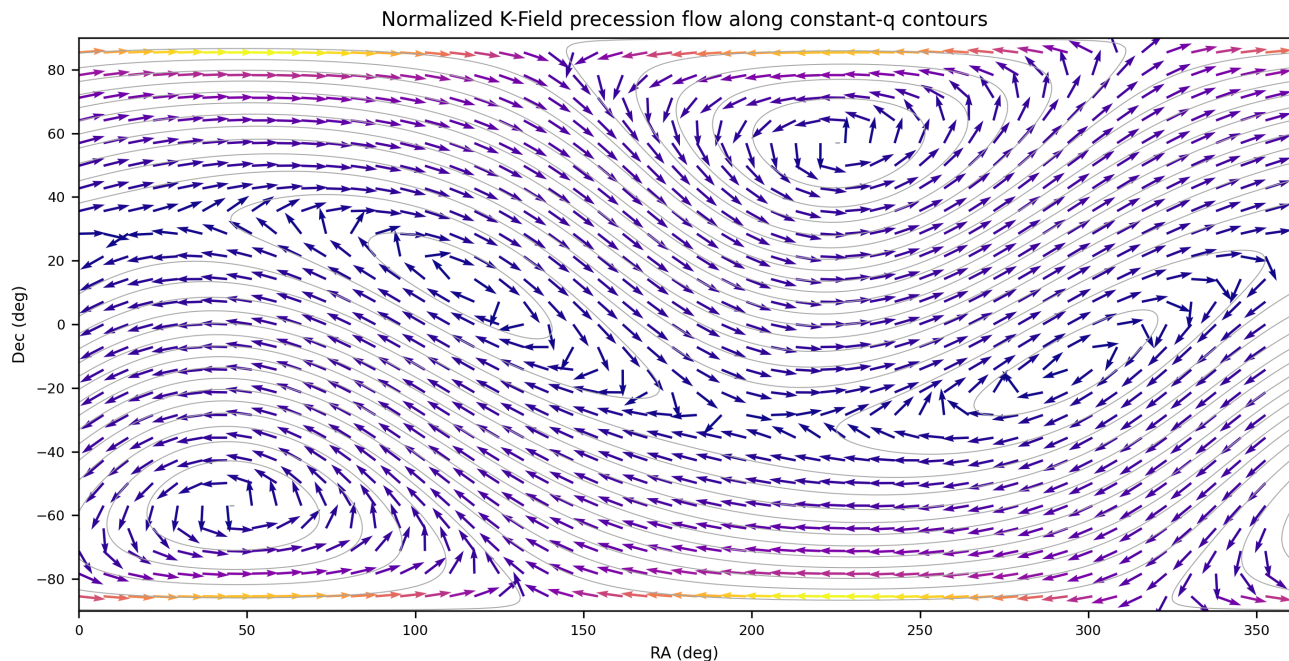


Figure 12. Orientation-space precession field generated from the exact G_{phase} matrix. The flow is tangent to the cost isolines and vanishes at the eigenaxes.

11.2 Dwell-time weighting

A mixed ensemble sampled at one time accumulates in slow sections of an orbit. The first-order residence density along an arc is proportional to $1/||n_{\text{dot}}||$. This weight is distinct from path multiplicity: Ω_N counts discrete propagation histories reaching an endpoint, while dwell time measures how long orientation flow remains near a direction.

$$\begin{aligned} \text{snapshot density along one mixed orbit proportional to } dt/ds \\ = 1 / ||n_{\text{dot}}|| \end{aligned}$$

12. Curvature organization beyond scalar cost and speed

Equal phase cost does not imply equal contour spacing, turning rate, tangent curvature, or focusing of neighboring trajectories. The local derivative hierarchy therefore evaluates slope magnitude and the two signed curvature coordinates of the slope-magnitude surface. The corpus occupancy analysis shows that curvature carries independent organization after conditioning jointly on phase cost and slope magnitude.

Test	Statistic	Degrees of freedom	p value
Signed RA curvature deciles	G=22.9474	-	0.00632
Signed Dec curvature deciles	G=34.8533	-	6.324e-05
Full cubic vs additive cubic	Delta G=79.5962	3	3.747e-17
Curvature added after slope magnitude	Delta G=65.0276721	10	4.00419e-10
Curvature added after q and slope	Delta G=32.9748532	8	6.22341e-05
Covariant east-curvature deciles	G=70.8216	-	1.0506e-11

The independent curvature contribution means that occupancy cannot be reconstructed from a Boltzmann-like phase-cost factor alone. It requires the path-fiber measure, the cost weighting, the precessional residence factor, and local focusing or defocusing.

13. Unified K-Field population measure

13.1 Three native weighting layers

The complete population measure contains three structurally different factors. Path multiplicity counts how many admissible histories arrive. Phase selection weights the directional transition cost. Residence and curvature weight how orientation flow occupies the local phase portrait.

$$\begin{aligned} P_N(n) \text{ proportional to } \Omega_N(n) \\ \times \exp[-\beta_K q(n)] \\ \times C(n) / ||n \times G_{\text{phase}} n|| \end{aligned}$$

$C(n)$ denotes the measured local curvature-focusing contribution. It is not a replacement for the other factors. Each factor answers a different question: how many paths arrive, how costly the direction is, and how long or how strongly the flow occupies the direction.

13.2 Endpoint certainty plus propagation entropy

The unified measure does not weaken endpoint determinism. The scalar address still identifies one exact location. Statistical behavior enters through the multiplicity and transport of admissible histories over the explicit address space.

14. Symmetric path resolution on a curved cost surface

14.1 Equal path length does not imply equal energy change

Consider a local coordinate s on the phase-cost surface and two admissible paths of equal length ℓ , one toward $s+\ell$ and one toward $s-\ell$. Equal path probability makes the mean spatial increment zero. The associated phase-cost increments need not cancel because the surface may bend between the two destinations.

$$E[\Delta C] = [C(s+\ell) + C(s-\ell)]/2 - C(s)$$

Taylor expansion separates slope from curvature. The first derivative changes sign between the two paths and cancels. The second derivative has the same sign for both paths and survives. All odd derivatives cancel under an exactly symmetric pair.

$$C(s \pm \ell) = C(s) \pm \ell C_{\text{prime}}(s) + \ell^2 C_{\text{second}}(s)/2 \pm \ell^3 C_{\text{third}}(s)/6 + \dots$$

$$E[\Delta C] = \ell^2 C_{\text{second}}(s)/2 + O(\ell^4)$$

For a quadratic cost cross-section, every derivative above second order vanishes. The curvature law is then exact rather than approximate. A locally linear surface has zero second derivative and produces no mean energy change. Convex and concave sections produce opposite signs.

Local geometry	Second derivative	Mean equal-path cost change
Linear	$C_{\text{second}}=0$	$E[\Delta C]=0$
Convex	$C_{\text{second}}>0$	$E[\Delta C]>0$
Concave	$C_{\text{second}}<0$	$E[\Delta C]<0$

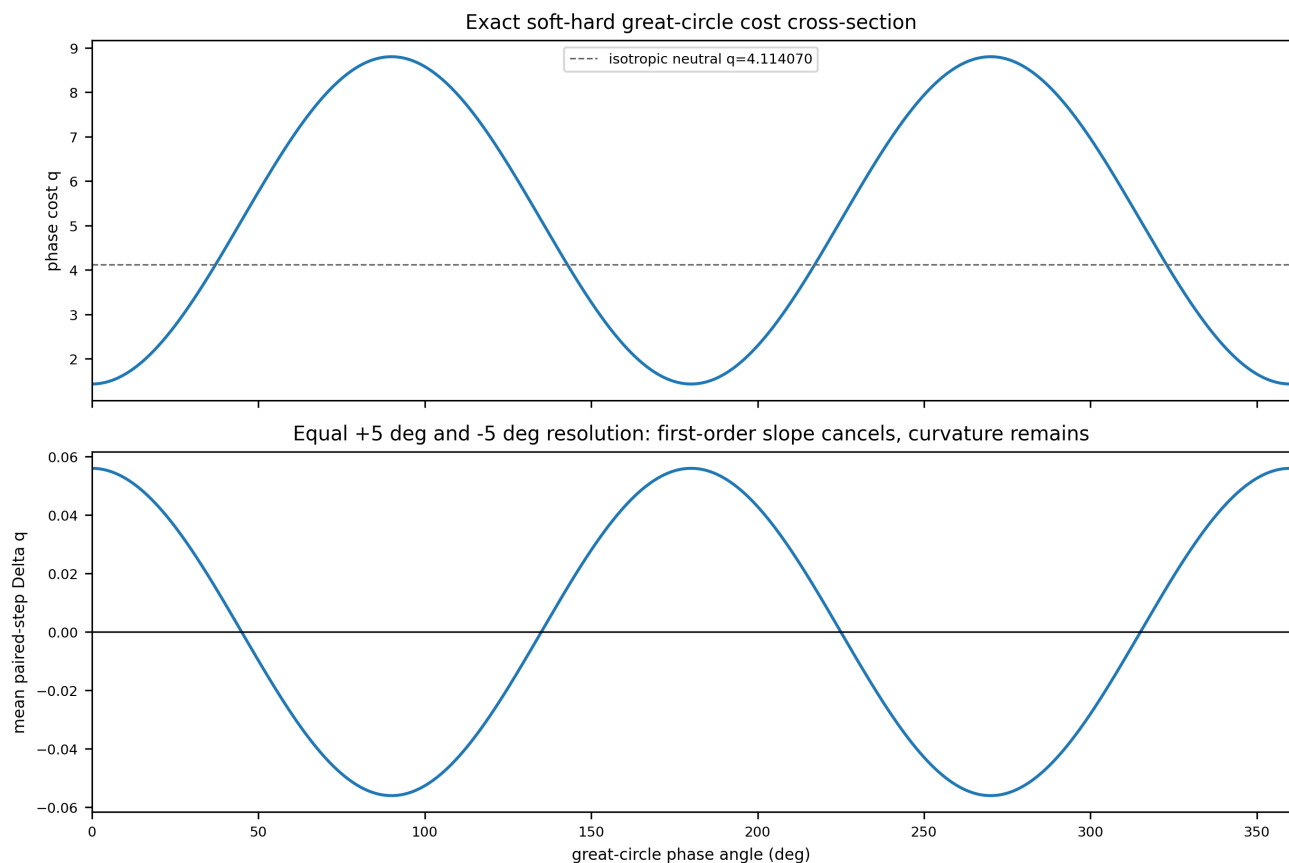


Figure 14. Exact paired-step rectification on the phase soft-hard great circle. The two paths have equal angular length and equal probability. Their average phase-cost change is nonzero wherever the cross-section is curved.

15. Multidimensional Hessian-covariance law

15.1 Zero mean displacement and nonzero second moment

Let δx be a local K-Field path increment. Equal forward and reverse probabilities give $E[\delta x]=0$. They do not eliminate the second moment $\Sigma=E[\delta x \delta x^T]$. The path entropy is the combinatorial source of this nonzero covariance: many admissible resolutions remain even after the endpoint description is explicit.

$$\begin{aligned} E[\delta x] &= 0 \\ \Sigma &= E[\delta x \delta x^T] \neq 0 \end{aligned}$$

15.2 Exact quadratic contraction

For a smooth local cost $q(x)$, expand through second order. The gradient term averages to zero, while the Hessian contracts with the path covariance. For a quadratic cost, the formula is exact.

$$q(x+\delta x) = q(x) + \text{grad}(q)^T \delta x + \delta x^T H_q \delta x / 2 + \dots$$

$$E[\Delta q] = \text{trace}(H_q \Sigma)/2$$

This equation is the local automatic pumping law. H_q records how the phase-cost surface bends. Σ records how the equal-weight path ensemble spreads. Their contraction determines the signed mean phase-energy change.

15.3 Exact one-event Cell3 covariance

The unbiased primitive alphabet contains the six physical increments $\pm a$, $\pm b$, and $\pm c$, each with probability $1/6$. In address coordinates the one-event covariance is $I/3$. In physical ICRS coordinates it is $BB^T/3$.

$$\begin{aligned} \Sigma_{\text{address}} &= I/3 \\ \Sigma_r &= B B^T / 3 \text{ (nm}^2\text{)} = \\ &\begin{bmatrix} 0.036202643451 & -0.002235261220 & -0.008210256635 \\ -0.002235261220 & 0.065976551592 & 0.052473610288 \\ -0.008210256635 & 0.052473610288 & 0.258928938414 \end{bmatrix} \end{aligned}$$

At physical radius R and direction n , a small displacement changes orientation by δn approximately equal to $P_n \delta r/R$, where $P_n = I - nn^T$ projects into the tangent plane. The corresponding angular covariance is $P_n \Sigma_r P_n / R^2$. This provides the direct bridge from the exact Cell3 path alphabet to curvature rectification on orientation space.

16. Exact curvature rectification on the quadratic orientation sphere

16.1 Equal geodesic paths

Let n be a unit orientation and u a unit tangent direction with $u^T n=0$. Two equal angular paths of length θ terminate at $n_{\text{plus}}=\cos(\theta)n+\sin(\theta)u$ and $n_{\text{minus}}=\cos(\theta)n-\sin(\theta)u$. These endpoints have equal path length and equal probability.

$$\begin{aligned} n_{\text{plus}} &= \cos(\theta)n + \sin(\theta)u \\ n_{\text{minus}} &= \cos(\theta)n - \sin(\theta)u \end{aligned}$$

Substitution into $q(n)=n^T G_{\text{phase}} n$ cancels the cross terms in the two-path mean and gives an exact finite-step result.

$$\begin{aligned} [q(n_{\text{plus}})+q(n_{\text{minus}})]/2 &= \cos^2(\theta) q(n) + \sin^2(\theta) u^T G_{\text{phase}} u \\ E[\Delta q] &= \sin^2(\theta)[u^T G_{\text{phase}} u - q(n)] \end{aligned}$$

16.2 Spherical tangent Hessian

The bracketed quantity is one half of the spherical Hessian in direction u . For small θ , $\sin(\theta)$ is approximately θ and the familiar Hessian-covariance law is recovered.

$$\text{Hess_S2 } q(u, u) = 2[u^T G_{\text{phase}} u - q(n)]$$

$$E[\Delta q] = \theta^2 \text{ Hess_S2 } q(u, u)/2 + O(\theta^4)$$

The exact formula shows that the current is intrinsic to the curved orientation surface. It is not created by unequal path lengths, unequal path probabilities, measurement error, or a hidden route register.

17. Soft, saddle, and hard phase axes

17.1 Eigenaxis directional coefficients

At a phase eigenaxis e_i with eigenvalue λ_i , a tangent path toward eigenaxis e_j produces the exact coefficient $\lambda_j - \lambda_i$. The sign is therefore determined directly by the eigenvalue ordering.

Starting axis	Tangent direction	Coefficient $E[\Delta q]/\sin^2(\theta)$	Interpretation
soft	toward saddle	0.683216272960	mean phase cost increases
soft	toward hard	7.371007681178	mean phase cost increases
saddle	toward soft	-0.683216272960	mean phase cost decreases
saddle	toward hard	6.687791408218	mean phase cost increases
hard	toward soft	-7.371007681178	mean phase cost decreases
hard	toward saddle	-6.687791408218	mean phase cost decreases

Every tangent direction at the soft axis points toward a higher eigenvalue, so symmetric resolution pumps phase cost upward. Every tangent direction at the hard axis points toward a lower eigenvalue, so symmetric resolution pumps phase cost downward. At the saddle, one tangent family pumps upward and the conjugate family pumps downward.

17.2 Saddle selection by path anisotropy

In the saddle tangent plane, let ψ measure direction from the soft eigenaxis toward the hard eigenaxis. The coefficient is $\lambda_{\text{soft}} \cos^2(\psi) + \lambda_{\text{hard}} \sin^2(\psi) - \lambda_{\text{saddle}}$. It crosses zero at $\psi = 17.725064614$ degrees and the symmetry-related directions. Therefore the native directional distribution of admissible paths selects the sign of the saddle current.

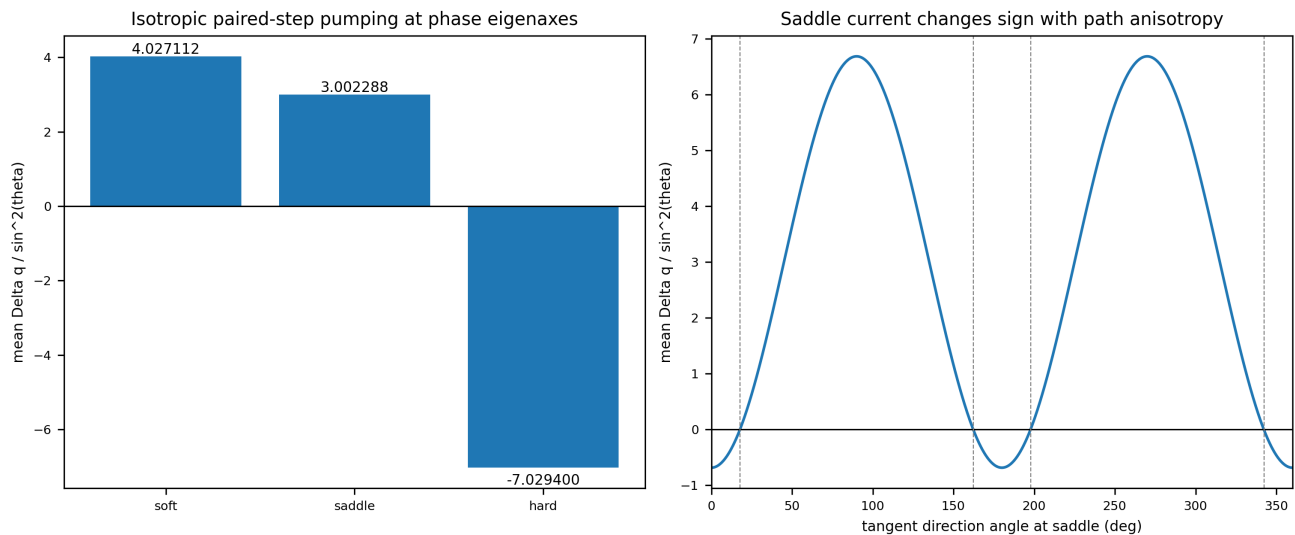


Figure 15. Left: exact isotropic phase-energy pumping coefficients at the soft, saddle, and hard axes. Right: exact directional coefficient in the saddle tangent plane, showing that path anisotropy selects the sign of the current.

18. Isotropic path resolution and the neutral-current contour

18.1 Tangent-direction average

When the local path ensemble is isotropic within the two-dimensional tangent plane, the mean of $u^T G_{\text{phase}} u$ is one half of the trace of G_{phase} restricted to that plane. Since the normal contribution is $q(n)$, the exact equal-pair mean becomes

$$E[\Delta q]_{\text{iso}} = \sin^2(\theta)[\text{trace}(G_{\text{phase}}) - 3q(n)]/2$$

The neutral cost is $\text{trace}(G_{\text{phase}})/3 = 4.114070071533$. Directions below this value have positive isotropic phase-energy pumping. Directions above it have negative pumping. The zero-current set is an exact projective contour on the quadratic surface.

Phase axis	q	$R_{\text{iso}} = E[\Delta q]/\sin^2(\theta)$	Current sign
soft	1.429328753487	4.027111977069	upward
saddle	2.112545026447	3.002287567629	upward
hard	8.800336434665	-7.029399544698	downward

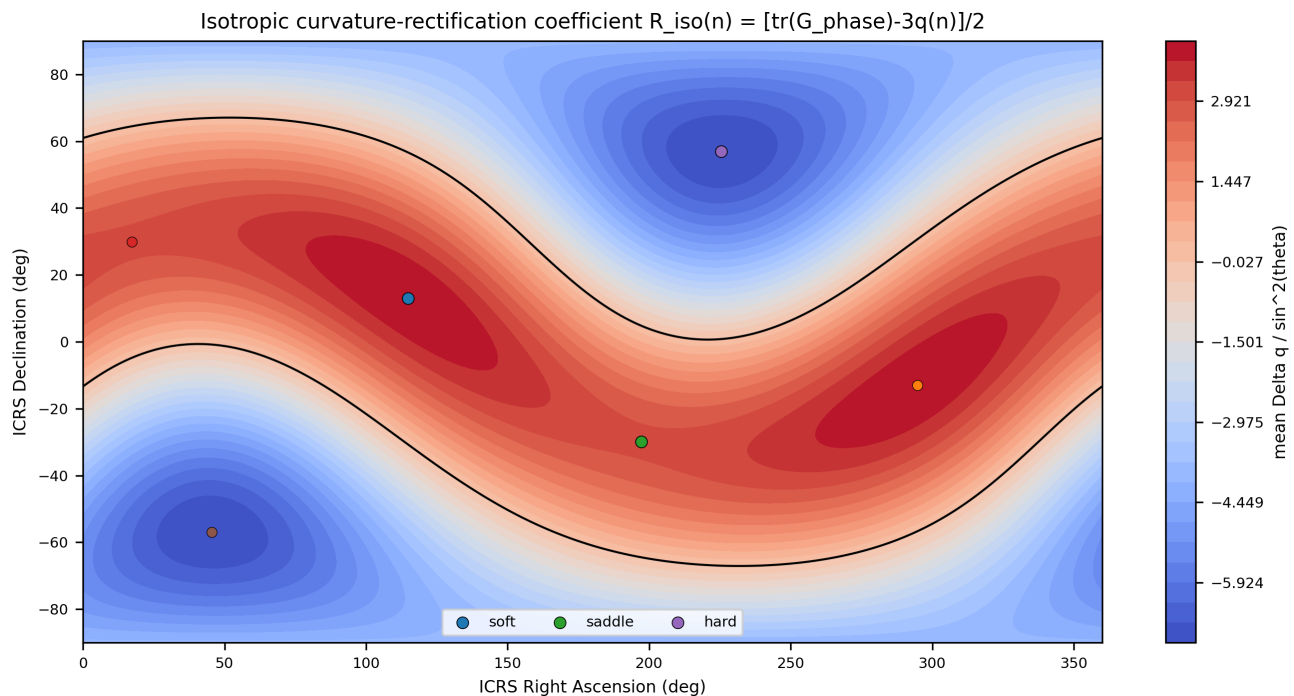


Figure 16. Exact projective map of the isotropic curvature-rectification coefficient. The black contour is the zero-current boundary $q = \text{tr}(G_{\text{phase}})/3$. Positive regions pump mean phase cost upward; negative regions pump it downward.

The unbiased local path ensemble therefore drives expected phase cost toward the neutral level rather than remaining stationary at every point. This is a curvature effect: the mean position increment remains zero.

19. Entropy-pumped phase-energy and population currents

19.1 Local phase-energy pumping rate

For event interval Δt , define the signed local phase-energy current as the expected phase-cost change per interval. This current may be nonzero even when the mean path displacement is zero.

$$j_q(x) = E[\Delta q]/\Delta t \\ = \text{trace}(H_q \Sigma)/(2 \Delta t)$$

This is curvature rectification of native K-Field entropy. Entropy supplies a population of equal-weight path resolutions and therefore a covariance. Curvature converts that symmetric covariance into a signed mean energy change.

19.2 Population current from the cost-entropy action

The local energy pumping and the redistribution of endpoint probability are related but distinct. The effective action is $A = \beta_K q - S_K/k_B$. A diffusion coefficient D converts its gradient into a population current.

$$\begin{aligned} J &= -D[\text{grad } p + p \text{ grad } A] \\ &= -D \text{ grad } p - D \beta_K p \text{ grad } q + (D/k_B) p \text{ grad } S_K \end{aligned}$$

The first term is ordinary spreading of endpoint probability. The second moves probability down the phase-cost gradient. The third moves probability up the entropy gradient toward states supplied by more admissible paths. Curvature rectification is the microscopic energy-pumping mechanism; the cost-entropy gradient is the macroscopic organizing field.

19.3 Coupling to cost-conserving precession

The previously defined precession flow follows constant- q contours, while the rectification field measures the signed mean change created by local stochastic resolution. These are complementary components of the complete dynamics: tangential precession transports orientation along the contour, and entropy-pumped curvature current changes the mean phase-energy budget of the resolved path ensemble.

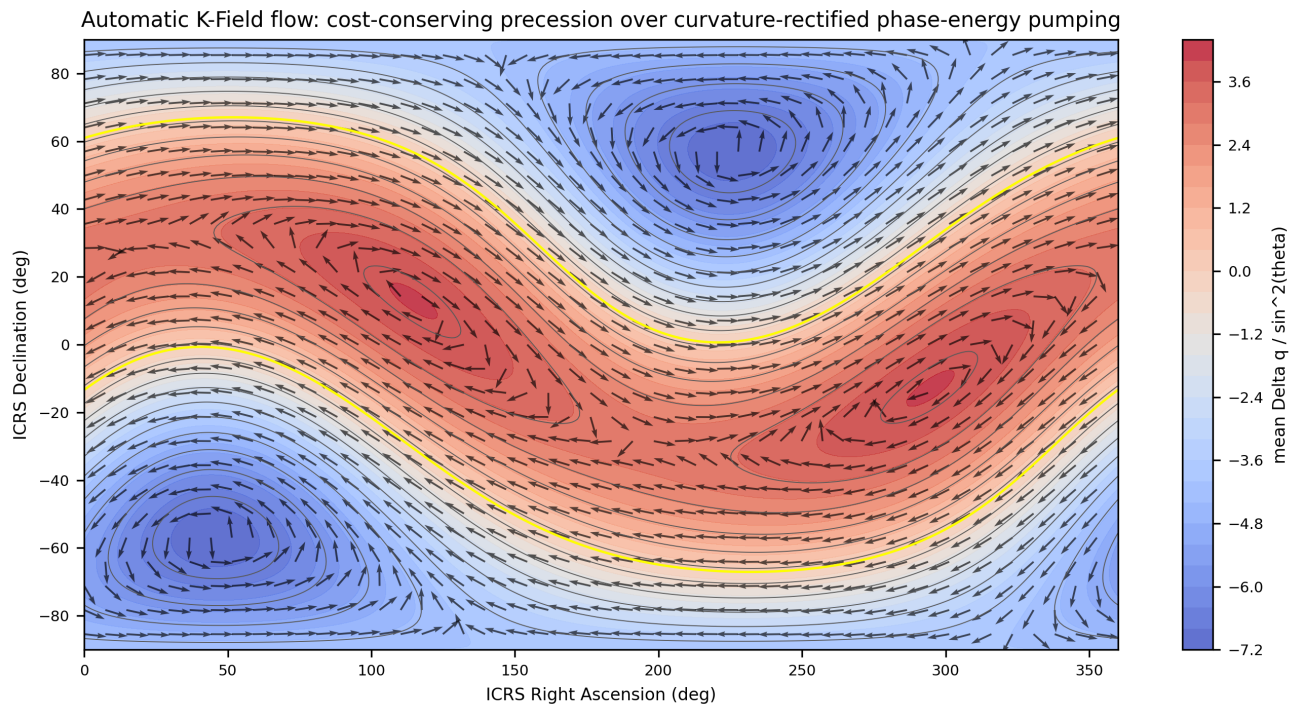


Figure 17. Exact phase portrait combining the cost-conserving precession direction with the curvature-rectified mean phase-energy change. The arrows show tangential orientation flow; the field colors show local upward or downward phase-energy pumping under symmetric path resolution.

The central mechanism is therefore exact: zero-mean path uncertainty plus nonzero phase curvature produces nonzero mean phase-energy flow. A locally linear cost has no rectification. Convex, concave, and saddle regions produce positive, negative, and path-anisotropy-selected currents respectively.

20. Central findings

Finding	Exact or derived result
Explicit endpoint address	$\rho(k)$ uniquely identifies k and therefore B_k , λ , polarity, and recursive digits.
Native path entropy	$S_N(k) = k_B \ln \Omega_N(k)$ for equal-weight admissible histories.
10-event history count	60,466,176 exact ordered histories and 891 reachable endpoints.
Entropy decomposition	$17.9175946923 = 5.3647281883 + 12.5528665040$ nats.
Path fraction of information	70.0588818957 percent remains after the endpoint is known.
Phase cost	$q(n) = n^T G_{\text{phase}} n$ with eigenvalues 1.4293287535, 2.1125450264, and 8.8003364347.
Entropy surface	$q_S(n) = n^T B^*(-T) B^*(-1) n$ with eigenvalues 1.2228728514, 6.3334327140, and 9.2862895511 nm^{-2} .
Independence in unbiased propagation	$\text{corr}(q, H_{\text{fiber}}) = -0.03445$; probability-weighted $\text{corr} = 0.00137$.
Cost-entropy action	$A_N(k) = \beta_K q(k) - \ln \Omega_N(k)$.
Effective orientation matrix	$G_{\text{eff}} = \beta_K G_{\text{phase}} + [3R^2/(2N)] G_S$.
Non-coaxial competition	Changing the entropy-to-cost ratio rotates the preferred direction.
Complete occupancy	Path multiplicity $\times$ phase weight $\times$ dwell/focusing weight.
Curvature rectification	$E[\Delta q] = \text{trace}(H_q \Sigma)/2$; symmetric paths cancel slope but retain curvature.
Neutral phase cost	$q_{\text{neutral}} = \text{tr}(G_{\text{phase}})/3 = 4.114070071533$.
Eigenaxis pumping	Soft and saddle are positive under isotropic resolution; hard is negative.
Automatic phase-energy current	$j_q = E[\Delta q]/\Delta t$ without a nonzero mean displacement.
Population current	$J = -D \text{grad } p - D \beta_K p \text{grad } q + (D/k_B) p \text{grad } S_K$.

21. New pathways to discovery and direct tests

21.1 Event-count scaling

The entropy coefficient $3R^2/(2N)$ decreases as event count increases at fixed physical radius. The phase term does not carry this N dependence. The effective eigensystem should therefore evolve systematically with propagation depth. Measuring this rotation provides a direct separation of phase-cost and entropy contributions.

21.2 Radius scaling

At fixed event count, the entropy penalty grows as R^2 . Distant endpoints require increasingly specific address histories and therefore display stronger directional entropy anisotropy. A radius-resolved occupancy map can test the G_S surface independently of β_K .

21.3 Controlled transition bias

The exact 10-event baseline is unbiased. Introducing a measured directional phase transition law into the six primitive probabilities changes $\Omega_N(k)$ into a weighted path partition sum. Comparing unbiased and phase-weighted runs isolates the direct contribution of G_{phase} while preserving the explicit endpoint address.

21.4 Curvature-conditioned entropy test

Endpoints may be binned jointly by q , q_S , slope magnitude, and curvature state. If the unified measure is complete, adding q_S or exact finite- $N \ln \Omega$ should improve occupancy prediction after conditioning on q and derivatives. The improvement must use one common conversion law across all bins.

21.5 Recursive Lucas and Supergroup extension

The scalar address and balanced-ternary carry extend to Lucas Groups and Lucas Supergroups without changing the endpoint theorem. Multi-path entropy can therefore be computed at each recursive depth. The phase matrix remains locked while the path partition expands. This produces a direct program for identifying entropy-driven band populations and closure transitions across the complete hierarchy.

21.6 Curvature-reversal experiment

Select orientations on opposite sides of the neutral contour while holding event count and local path covariance fixed. The model predicts a sign reversal of the mean phase-energy change. A locally linear calibration segment predicts zero. The comparison isolates curvature rectification from slope bias.

21.7 Saddle anisotropy experiment

At the phase saddle, rotate an otherwise matched path-covariance ellipse through the tangent plane. The current must change sign at the calculated zero directions. This is a direct test that the Hessian and covariance contract as a tensor rather than through a scalar cost alone.

21.8 Event-rate scaling

At fixed one-event covariance, the phase-energy pumping rate scales with event frequency through $j_q = E[\Delta q]/\Delta t$. Separately varying event frequency and path step size distinguishes the temporal rate from the quadratic step-size dependence.

22. Reference formulas

Concept	Formula
Physical displacement	$r(k) = Bk$
Direct metric distance	$\ r\ ^2 = k^T G k$
Signed scalar address	$\rho(k) = \sigma(k) \sqrt{k^T G k}$
Endpoint direction	$n(k) = Bk / \sqrt{k^T G k}$
Phase cost	$q(n) = n^T G_{\text{phase}} n$
Address-space phase cost	$q(k) = 4(k^T W k) / (k^T G k)$
Path multiplicity	$\Omega_N(k) = \Gamma_N(k) $
Equal-weight path entropy	$S_N(k) = k_B \ln \Omega_N(k)$
Shortest-path multiplicity	$\Omega_{\min} = N_{\min}! / (k_a ! k_b ! k_c !)$
History generating function	$\Omega_N(k) = [z^a k] (z_a + z_a^{-1} + z_b + z_b^{-1} + z_c + z_c^{-1})^N$
Entropy matrix	$G_S = B^T(-T)B^A(-1)$
Large-N entropy	$\ln \Omega_N(R, n) = \text{constant} - [3R^2/(2N)] n^T G_S n + \dots$
Endpoint weight	$W_N(k) = \Omega_N(k) \exp[-\beta_K q(k)]$
Effective action	$A_N(k) = \beta_K q(k) - \ln \Omega_N(k)$
Effective matrix	$G_{\text{eff}} = \beta_K G_{\text{phase}} + [3R^2/(2N)] G_S$
Precession	$\dot{n} = -(2/L) n \times G_{\text{phase}} n$
Cost conservation	$dq/dt = 0$
Dwell time	density proportional to $1/ n_{\text{dot}} $
Unified population	P proportional to $\Omega_N \exp[-\beta_K q] C / n \times G_{\text{phase}} n $
Symmetric paired step	$E[\Delta C] = [C(s+ell) + C(s-ell)]/2 - C(s)$
Quadratic curvature exact law	$E[\Delta C] = ell^2 C_{\text{second}}/2$
Hessian-covariance law	$E[\Delta q] = \text{trace}(H_q \Sigma)/2$
Exact spherical pair	$E[\Delta q] = \sin^2(\theta) [u^T G_{\text{phase}} u - q(n)]$
Spherical Hessian	$\text{Hess}_S q(u, u) = 2[u^T G_{\text{phase}} u - q(n)]$
Isotropic rectification	$E[\Delta q]_{\text{iso}} = \sin^2(\theta) [\text{tr}(G_{\text{phase}}) - 3q]/2$
Neutral cost	$q_{\text{neutral}} = \text{tr}(G_{\text{phase}})/3$
Phase-energy current	$j_q = E[\Delta q]/\Delta t$
Population current	$J = -D[\text{grad } p + p \text{ grad}(\beta_K q - S_{K/B})]$

Appendix A. Complete 27-node Cell3 ledger

Address	lambda	polarity	kind	x (nm)	y (nm)	z (nm)	rho native
(-1, -1, -1)	-3	-1	corner	-0.176351350144	0.118682429904	0.637311498326	-0.041809908117110
(-1, -1, 0)	-2	+1	edge center	-0.181416008957	-0.076211421278	0.292444294144	-0.021936074880302
(-1, -1, 1)	-1	-1	corner	-0.186480667770	-0.271105272459	-0.052422910037	-0.020735999640663
(-1, 0, -1)	-2	+1	edge center	-0.158685962221	0.222342355125	0.363610955630	-0.028302770281269
(-1, 0, 0)	-1	-1	face center	-0.163750621035	0.027448503943	0.018743751448	-0.010398529391216
(-1, 0, 1)	0	+1	edge center	-0.168815279848	-0.167445347239	-0.326123452733	-0.025117311897496
(-1, 1, -1)	-1	-1	corner	-0.141020574299	0.326002280346	0.089910412934	-0.022802164337825
(-1, 1, 0)	0	+1	edge center	-0.146085233113	0.131108429164	-0.254956791248	-0.020024508730742
(-1, 1, 1)	1	-1	corner	-0.151149891926	-0.063785422018	-0.599823995429	-0.038700006936387
(0, -1, -1)	-2	+1	edge center	-0.012600729109	0.091233925961	0.618567746877	-0.038919790124290
(0, -1, 0)	-1	-1	face center	-0.017665387922	-0.103659925221	0.273700542696	-0.018247102565364
(0, -1, 1)	0	+1	edge center	-0.022730046735	-0.298553776403	-0.071166661485	-0.019152826848624
(0, 0, -1)	-1	-1	face center	0.005064658813	0.194893851182	0.344867204181	-0.024654282949930
(0, 0, 0)	0	+1	body center	0.000000000000	0.000000000000	0.000000000000	0.000000000000000
(0, 0, 1)	1	-1	face center	-0.005064658813	-0.194893851182	-0.344867204181	0.024654282949930
(0, 1, -1)	0	+1	edge center	0.022730046735	0.298553776403	0.071166661485	0.019152826848624
(0, 1, 0)	1	-1	face center	0.017665387922	0.103659925221	-0.273700542696	0.018247102565364
(0, 1, 1)	2	+1	edge center	0.012600729109	-0.091233925961	-0.618567746877	0.038919790124290
(1, -1, -1)	-1	-1	corner	0.151149891926	0.063785422018	0.599823995429	0.038700006936387
(1, -1, 0)	0	+1	edge center	0.146085233113	-0.131108429164	0.254956791248	0.020024508730742
(1, -1, 1)	1	-1	corner	0.141020574299	-0.326002280346	-0.089910412934	0.022802164337825
(1, 0, -1)	0	+1	edge center	0.168815279848	0.167445347239	0.326123452733	0.025117311897496
(1, 0, 0)	1	-1	face center	0.163750621035	-0.027448503943	-0.018743751448	0.010398529391216
(1, 0, 1)	2	+1	edge center	0.158685962221	-0.222342355125	-0.363610955630	0.028302770281269
(1, 1, -1)	1	-1	corner	0.186480667770	0.271105272459	0.052422910037	0.020735999640663
(1, 1, 0)	2	+1	edge center	0.181416008957	0.076211421278	-0.292444294144	0.021936074880302
(1, 1, 1)	3	-1	corner	0.176351350144	-0.118682429904	-0.637311498326	0.041809908117110

Appendix B. Six boundary-plane membership ledger

Channel	Face	Sign	Area (nm ²)	Opposite spacing (nm)	Nine member addresses
a	bc	-1	0.357809539128562	0.329546529709293	(-1, -1, -1), (-1, -1, 0), (-1, -1, 1), (-1, 0, -1), (-1, 0, 0), (-1, 0, 1), (-1, 1, -1), (-1, 1, 0), (-1, 1, 1)
a	bc	+1	0.357809539128562	0.329546529709293	(1, -1, -1), (1, -1, 0), (1, -1, 1), (1, 0, -1), (1, 0, 0), (1, 0, 1), (1, 1, -1), (1, 1, 0), (1, 1, 1)
b	ac	-1	0.261106235799531	0.451597379724130	(-1, -1, -1), (-1, -1, 0), (-1, -1, 1), (0, -1, -1), (0, -1, 0), (0, -1, 1), (1, -1, -1), (1, -1, 0), (1, -1, 1)
b	ac	+1	0.261106235799531	0.451597379724130	(-1, 1, -1), (-1, 1, 0), (-1, 1, 1), (0, 1, -1), (0, 1, 0), (0, 1, 1), (1, 1, -1), (1, 1, 0), (1, 1, 1)
c	ab	-1	0.194869165469900	0.605097741514743	(-1, -1, -1), (-1, 0, -1), (-1, 1, -1), (0, -1, -1), (0, 0, -1), (0, 1, -1), (1, -1, -1), (1, 0, -1), (1, 1, -1)
c	ab	+1	0.194869165469900	0.605097741514743	(-1, -1, 1), (-1, 0, 1), (-1, 1, 1), (0, -1, 1), (0, 0, 1), (0, 1, 1), (1, -1, 1), (1, 0, 1), (1, 1, 1)

Appendix C. Exact 10-event minimum-step shell ledger

Minimum steps	Endpoint count	History count	Probability	Mean distance (nm)	Min distance (nm)	Max distance (nm)
0	1	1172556	0.019391932441701	0.000000000000	0.000000000000	0.000000000000
2	18	14770980	0.244285003238836	0.910344607957	0.615518338108	1.584639530085
4	66	24737040	0.409105414570950	1.600727221980	0.855955284386	3.169279060171
6	146	15122430	0.250097343678555	2.297185737704	1.332794175875	4.753918590256
8	258	4203060	0.069510927894630	2.971265986868	1.702907109406	6.338558120341
10	402	460110	0.007609378175329	3.623015653745	2.140966059563	7.923197650427

Appendix D. Phase-cost decile and entropy ledger

Decile	Endpoints	Equal mean q	Equal mean H_fiber	Probability mass	Weighted mean q	Weighted mean H_fiber
1	90	1.807732030660	9.168856694238	0.126241057480	1.817453340855	12.620854167986
2	88	2.310675353217	8.164117643579	0.063335078441	2.259748028725	12.110991443466
3	90	2.926854443606	8.698677519493	0.111754479066	2.891086711316	12.603199696633
4	88	3.544988075327	7.766851497015	0.088773961826	3.654137034775	12.619317426742
5	90	4.154670230333	8.218215443782	0.106519453124	4.156990404452	12.646318191627
6	88	4.829133600660	8.837576688483	0.108703318695	4.822572356108	12.574495876907
7	90	5.682100429926	9.146510258624	0.101554859365	5.827192693245	12.471326584779
8	88	6.546308740629	9.059664669965	0.100539845616	6.514074479677	12.375918072542
9	90	7.440349792917	8.118601074233	0.071087511802	7.339358159362	12.196466992487
10	88	8.301490910307	7.933905948470	0.102098502144	8.324257722923	12.742960809481

Appendix E. Source and integrity ledger

Role	Source file	Epoch	SHA-256
serialized_address_space	K-Field_overdetermined_one-dimensional_serialized_spectral_address_space_1785179337(2).json	1785179337	34ccd7ee5c1e7a9071d4cf1ef7164620e067eba133fe33ce43044d0c1556e51b
exact_10_event_propagation	K-Field_10_event_Cell3_propagation_experiment_1785204765.json	1785204765	c446b6f9a9b246ad9e8aabe013a472b92eddbfdae6112210f4bc896cef3363f6
quadratic_phase_cost_and_precession	K-Field_galaxy_gyroscopic_precession_geometry_1784741310.json	1784741310	b78e56ea889d4f790243244d27eacce3d67efd2313a3d09953c22287a4f07249
phase_metric_and_address_coupling_identity	K-Field_single_cell3_lateral_force_reduction_1784985105.json	1784985105	62e4472aacc7df503cc4b5120d7abce97589891e8d22843177ee547e35b71a6f

All geometric figures were regenerated directly from the numerical matrices and address rules printed in this report. All endpoint statistics were computed from the complete exact destination registry. The machine-readable companion contains the full 891-endpoint cost-entropy ledger, the complete mixing sweep, curvature-rectification invariants, one-event covariance matrices, all node and plane records, and hashes of every generated figure.

Archival colophon

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